



DEALER BEST PRACTICE SERIES

WesTrac Apprentice Program

Application Maintenance Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

This Best Practice provides an overview of Westrac's Australian apprentice program. This curriculum was developed by WesTrac to address the specific requirements of Caterpillar machines and equipment.

The purpose of this best practice is to encourage dealer engagement with Caterpillar when developing an Apprenticeship Course. This document provides an example of Westrac's Apprentice Program.

WesTrac's apprentice program is a three and a half year course that is structured around Australian recognised training qualifications. The aim of this program is for WesTrac to educate and develop their employees through a combination of classroom style learning and hands on field experience. The implementation of this program has enabled and will continue to enable WesTrac to provide the best possible support and after sales service to its Caterpillar customers.

2.0 Best Practice Description

Apprentice Induction

All new apprentices are put through a one-week induction and are required to attend the following sessions:

- Management greeting
- Personal protective equipment issue
- Tool store operation
- Warranty and report writing introduction
- Clocking on/off procedure
- Signing of agreements
- Occupational Health and Safety (OH&S) training
- Human resources policies and procedures
- Drug and alcohol awareness
- Counselling availability

Further safety training is conducted during their first year after the induction period.

WesTrac Institute

The primary aim of the WesTrac Institute is to deliver a skilled and competent workforce.

Trained and competent personnel are essential when building a workplace culture that ensures:

- The highest priority on health and safety
- Applicable technical skills
- Robust processes and procedures
- Improved and standardised training at all employee levels

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WesTrac's Institute currently delivers training and assessment in the following areas:

- Certificate III in Automotive Mechanical Technology (Heavy)
- Post Trade Technical
- Machine Operation
- Occupational Safety and Health
- Training and Assessment

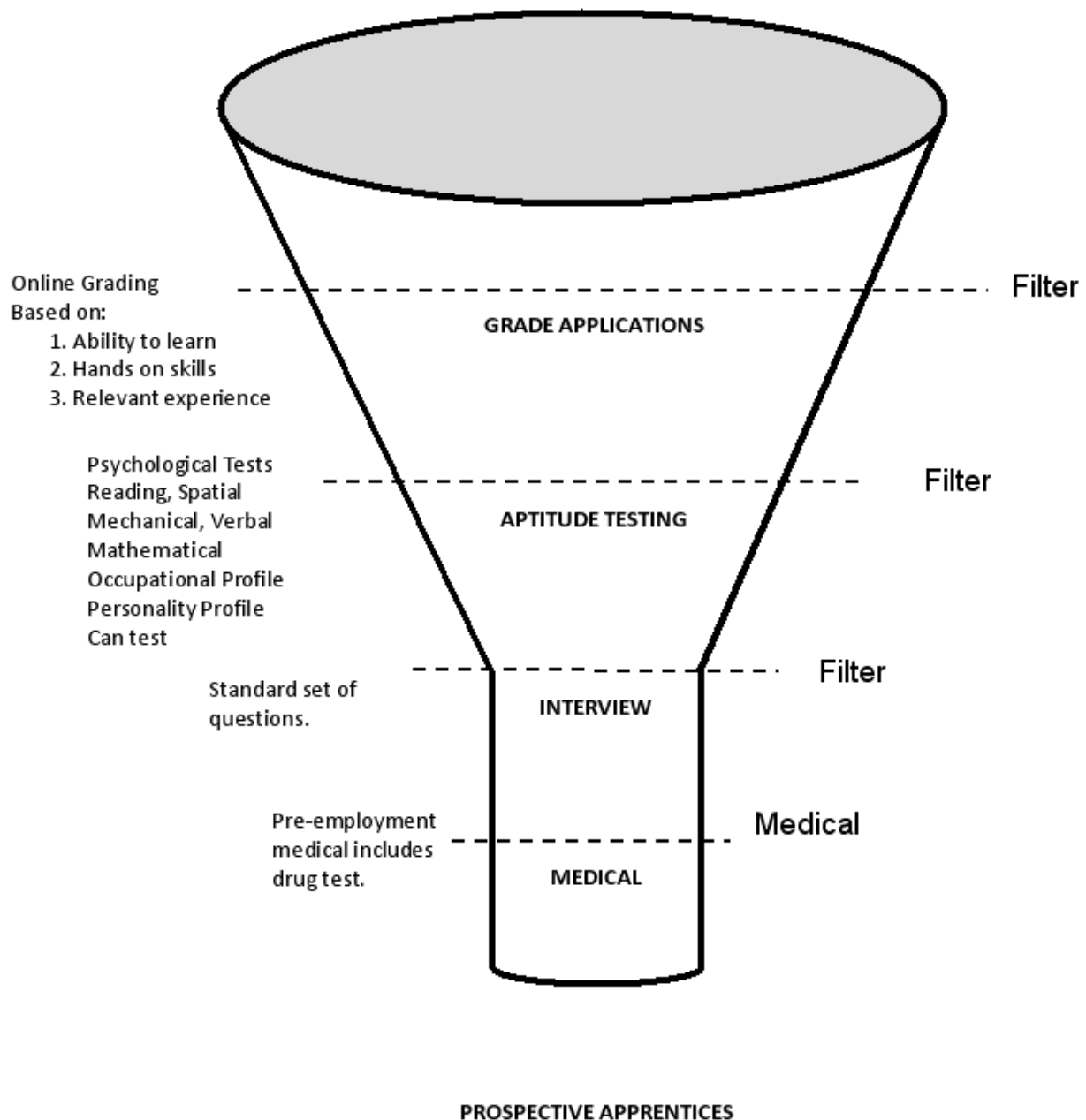
The WesTrac Institute curriculum has been developed through close consultation with various industries and the Western Australian government, ensuring apprentices receive a comprehensive trade qualification. This program is industry focussed and sets a new benchmark in the heavy mobile equipment mechanical trade.

The Institute provides an industry specific learning program that draws upon current and emerging industry technologies. This program also allows the immediate access to Caterpillar machines, components and simulators that will reinforce classroom learning.

The program exposes apprentices to technical, technological, business and interpersonal competence that is otherwise hard to come by.

WesTrac Institute trainers have recent and relevant industry experience and hold relevant training, assessment and occupational, safety and health qualifications.

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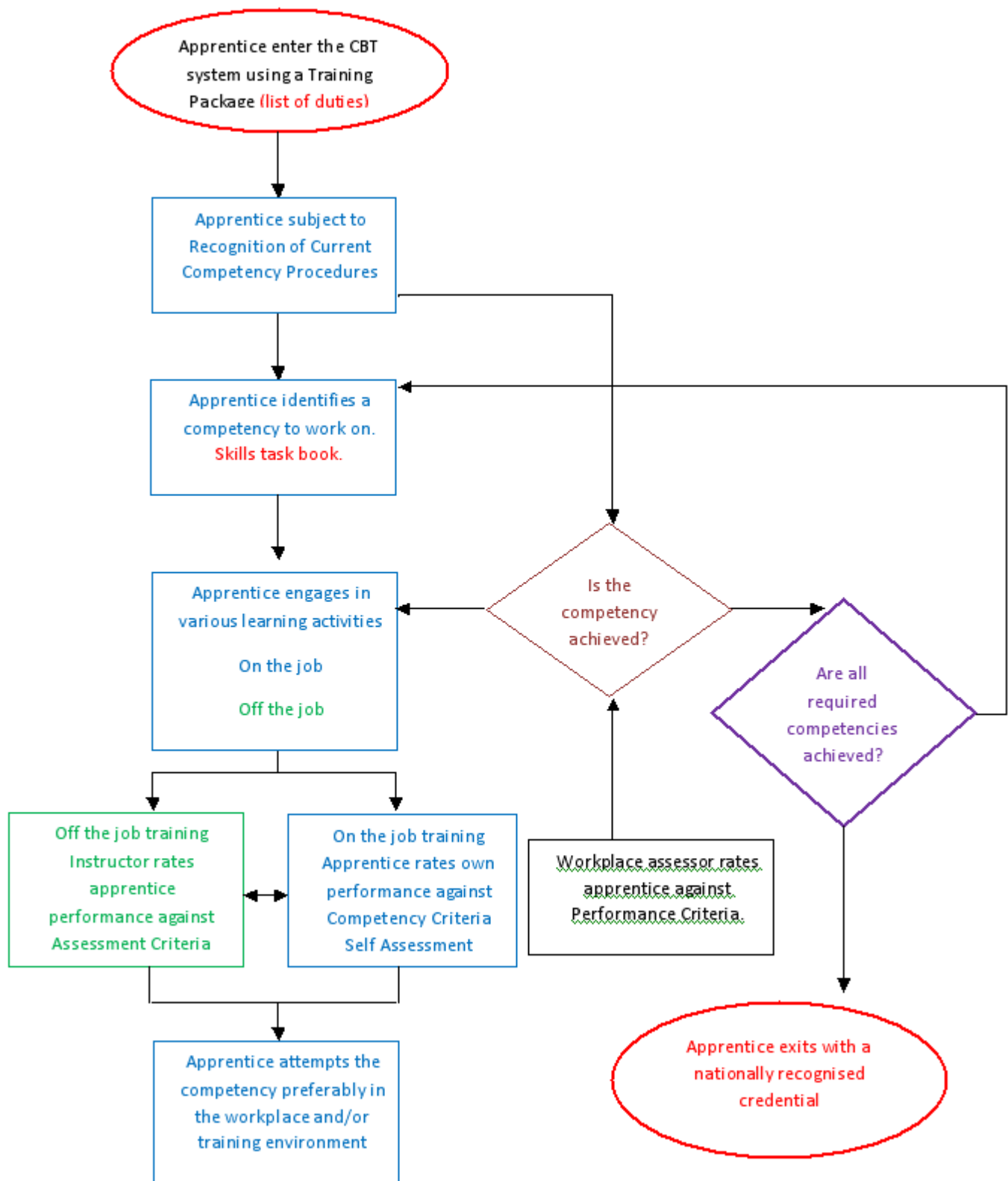
Apprentice Recruitment at WesTrac PTY Ltd

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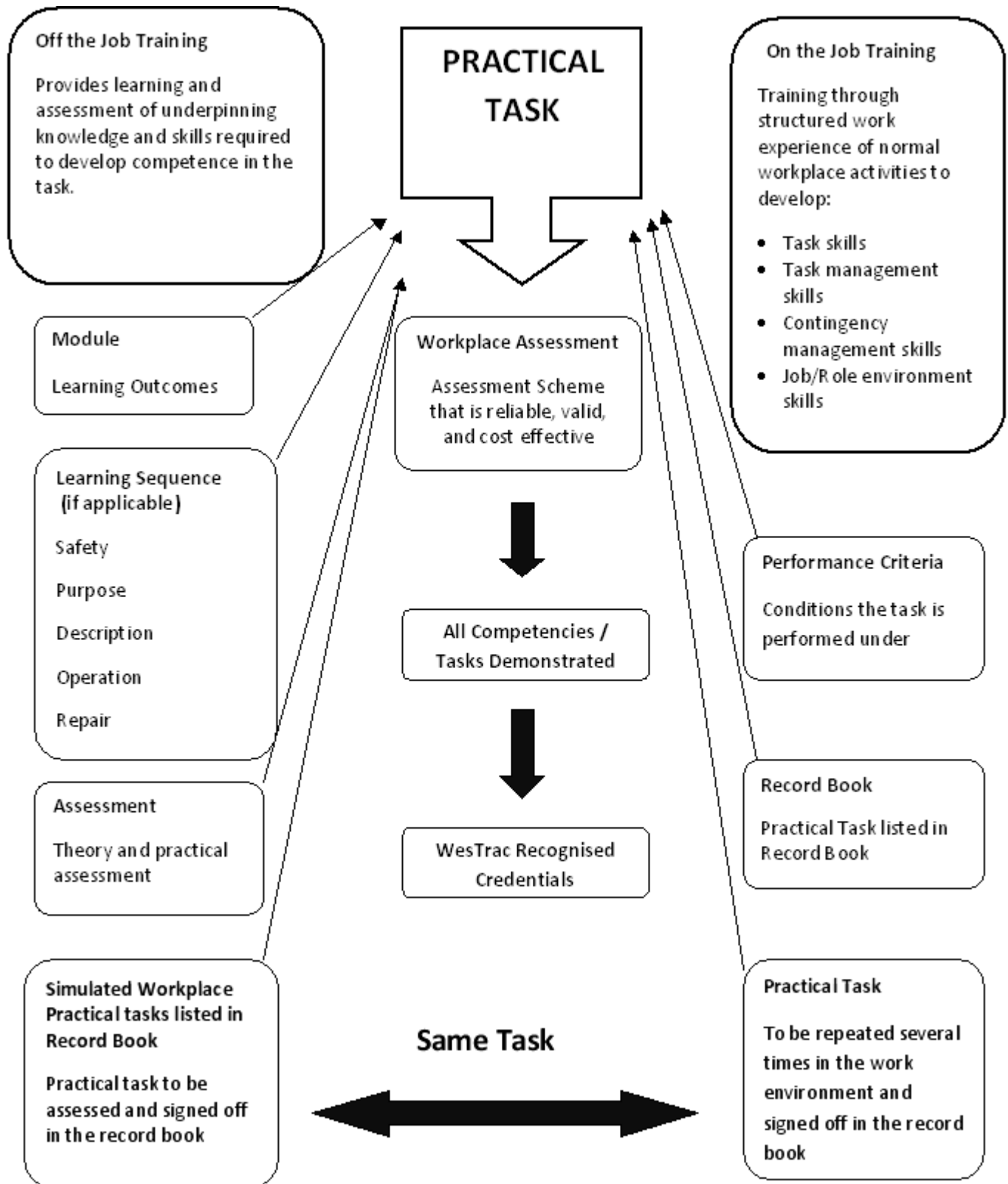
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Competency Based Training Pathway for Apprentice Training



WesTrac Apprentice Training Competency Based Training Model

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WesTrac workplace Assessor**Skills Testing**

- Select and create skills tests for apprentices that match the year of apprenticeship and expected skills level
- Carry out skills test – minimum four hours per apprentice year

Skills Testing Remedial

- Carry out remedial training (study, hands on)
- Retest skills that were below acceptable level

Competency Record Book

- Carry out regular audits on competency record book
- Liaise with Workshop Supervisor and Mentor to ensure competency tasks related to the particular section are available for the apprentice to complete

Competency Testing

- Monitor competency record book for completed tasks of a particular competency
- Carry out oral testing of that competency
- Carry out practical testing of competency or part of if necessary, to satisfy if participant is competent in the unit concerned

Off The Job Training

- Follow off the job training results
- Remedial coaching on the hold/s using teaching aids available

Tool Inspection

- Inspect tool kits on a regular basis to ensure that the kits meet the conditions of the WesTrac tool support contract

Monthly Assessments

- Ensure apprentices have the monthly assessments carried out and the assessment sheets are received for recording and filing. Poor performance areas in the assessment are to be followed up and addressed in the appropriate manner

Appearance

- Encourage apprentices to maintain a standard of dress and appearance that the company expects

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On The Job Training

- Cooperate with Supervisors and Mentors to raise the skill level of the apprentice in the section or area as quick as possible so as to be able to be in a position to carry out productive work
- Map out a chart of the skills required by the various sections and ensure that the apprentice is competent in these skills. This is achieved by working under suitable supervision or through simulated exercises.
- With the cooperation of the supervisor select suitable tasks within the section or area to match skills and ability of the apprentice.
- Have the apprentice carry out these tasks as a part of the apprentice duties thus allowing more qualified personnel to do more complex tasks.

Awards

- Encourage apprentice to excel in their given trades and compete in State and National skills competitions. Carry out skills coaching as directed.

Apprentice Initiatives

- No 1st Year apprentices in workshops
 - Dedicated practical skills areas (electrical laboratory, fabrication, hand skills focus);
- Fast track competent apprentices
- More branch, Field Service and MARC support
- Improve mentoring
- During their first year, apprentices attend dedicated skills areas placed on workshop floor
- Formal training in forklift operation, slinging and lifting is also provided
- The skill areas focus on: hand skills, fabrication and electrical
- Throughout their second year, apprentices are introduced to machine repairs and assembly
- The third year consists of component overhauling followed by field and regional sites
- Final year apprentices are made aware of the opportunities and support they can expect after completion of their apprenticeship.
- A minimum offering of two technical training programs per year ongoing
 - Career pathways into Service or Marketing occupations after completion of trade qualifications
 - Opportunity to attend Leadership and Management Training programs
 - High performing final year apprentices can be nominated to attend the Future Leaders Program

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WesTrac apprentice assessment

Workplace trainers are competent in the skills they teach using:

- Clearly stated skill requirements
- Skill-based observation of workplace performance
- Skill-based records of progress
- Skill-based criteria for certification
- Providing trainee access to records
- Immediate feedback on performance

Independent monitoring of training and assessment standards will assure valid, accurate and meaningful assessments

Learning strategy

To acquire new ***skills and knowledge*** on components or machines, the following sequence should be used.

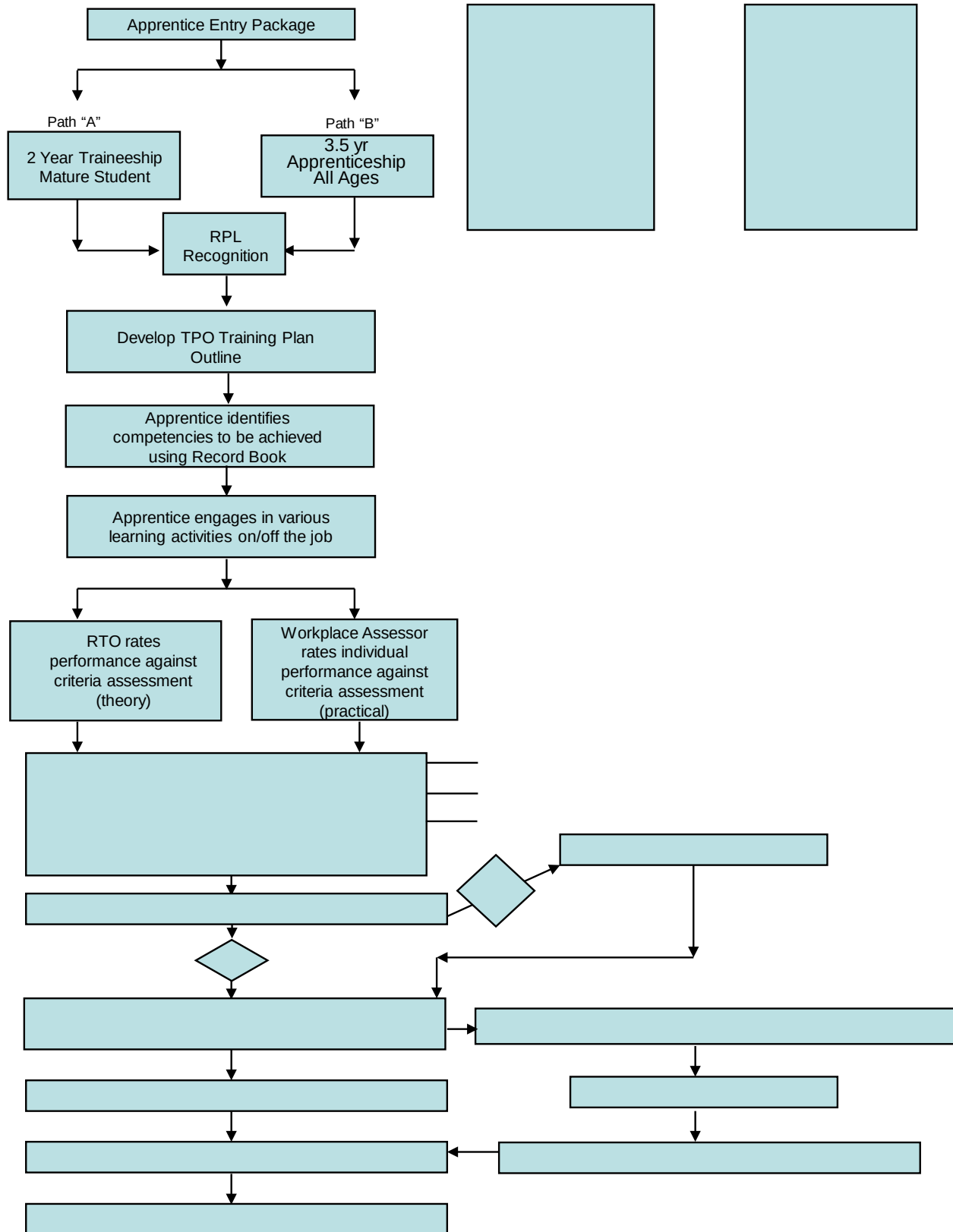
1. ***Occupational Safety and Health*** issues involved
2. ***Purpose*** of the component or machine
3. ***Description*** of the component or machine
4. ***Operation*** of the component or machine
5. ***Repair*** of the component or machine
6. ***Diagnosis*** of the component or machine
7. ***Failure Analysis*** of the component or machine

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Pathway for Apprentices / Traineeships



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Outcome: Heavy vehicle mechanical apprenticeship

On successful completion of this apprenticeship the participant will be able to:

- Describe and demonstrate safe work practices
- Maintain and operate hand and power tools and equipment typically involved with the earthmoving and industrial mechanic trade
- Identify earthmoving and industrial components and systems and state their purpose and operating principles
- Perform service, repair and overhaul procedures to earthmoving and industrial components/machines in accordance with manufactures or accepted industry specifications (performance and time factors)
- Apply diagnostic procedures to earthmoving and industrial components/machines using manufactures or accepted industry procedures (performance and time factors)

3.0 Implementation Steps

In conjunction with the Caterpillar/Dealer institute an apprentice program should be developed and tailored to the specific requirements of the dealer.

4.0 Benefits

Delivering outstanding value to customers, employees, shareholders and the community.

What the customer wants from today's service technician

- Only necessary work gets done
- Cost is reasonable
- Work be completed quickly
- Problem is fixed right the first time

The only way to satisfy this customer

- Quick, complete and accurate diagnosis

And the way to good diagnosis

- Be competent in the purpose, description and operation of the component/machine
- Discover the problem with the customer's help
- Write a clear and accurate repair order
- Test the diagnosis in a logical sequence

5.0 Resources Required

In order to implement an apprentice program a Caterpillar/Dealer institute will be required

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6.0 Supporting Attachments / References

Units of competence studied during the traineeship

AUR30405 – Certificate III in Automotive Mechanical Technology (Heavy)

Unit Number	Unit Description
AURC270103A	Apply Safe Working Practices
AURT200368A	Select and use bearings, seals, gaskets, sealants and adhesives
AURT225667A	Use & Maintain Measuring Equipment
AURT270278A	Use & Maintain Workplace Tools & Equipment
AURT271781A	Implement & Monitor Environmental regulations in the Automotive Mechanical Industry
AURE218676A	Test, service and charge batteries
AURE218708A	Carry out repairs to single electrical circuits
AURE224008A	Carry out soldering of electrical wiring/circuits
CBM9	Jacking & Blocking / Sling & Lifting
MEM5.7AB	Manual Heating & Thermal Cutting
AURT201170A	Inspect & Service Engines
AURT208170A	Service transmissions (hydrostatic)
AURT210170A	Inspect and service braking systems
AURT211170A	Inspect and service air braking systems
AURT310166A	Repair hydraulic braking systems
AURT311166A	Repair air braking systems
AURE319166A	Repair charging systems
AURE319266A	Repair starting systems
AURT301166A	Repair Engines & Associated Components
AURE318966A	Repair instruments and warning systems
AURE321771A	Service and repair electronic compression ignition engine management systems
AURE321271A	Service and repair electronic drive management systems
AURT366108A	Carry out diagnostic procedures
AURT215170A	Inspect and service steering systems
AURT401145A	Overhaul Engines & Associated Engine Components
AURT315166A	Repair steering systems
AURT202170A	Inspect & Service Cooling Systems
AURT202166A	Repair Cooling Systems
AURT203670A	Service Diesel Fuel Injection Systems
AURT303666A	Repair Diesel Fuel Systems
AURT305671A	Inspect & Repair Engine Forced Induction Systems
AURT222670A	Service air conditioning systems
CBHYD1	Hydraulic Fundamentals
AURT201164A	Remove and install engine assemblies
AURT304666A	Repair and replace emission control systems
AURT306666A	Repair transmissions (manual)
AURT209170A	Service Hydraulic Systems
AURT409166A	Repair Hydraulic Systems
AURT306170A	Inspect, service and/or repair clutch assemblies and associated operating
AURT207170A	Inspect & Service Transmissions (Automatic)
AURT307166A	Repair Transmissions (Automatic)
AURT208170A	Service Transmissions (Hydrostatic)
AURT212670A	Service Final Drives Assemblies
AURT312666A	Repair Final Drive Assemblies
AURT213170A	Service Final Drive (Driveline)
AURT313166A	Repair Final Drive (Driveline)
AURT216170A	Inspect & Service Suspension Systems
AURT316166A	Repair Suspension Systems

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AUR20705 – Certificate II in Automotive Mechanical (Transmissions)

Unit Number	Unit Description
AURT270278A	Use and maintain workplace tools and equipment.
AURT225667A	Use and maintain measuring equipment.
AURT271781A	Implement and monitor environmental regulations in the automotive mechanical industry.
AURC270103A	Apply safe working practices.
AURT213170A	Service final drive (driveline).
AURT207170A	Inspect and service transmissions (automatic).
AURT210170A	Inspect and service braking systems.
AURE218708A	Carry out repairs to single electrical circuits.
AURT202170A	Inspect and service cooling systems.
AURT206670A	Inspect and service transmissions (manual).
AURT212670A	Service final drive assemblies.
AURT307166A	Repair Transmissions (automatic)
AURE321271A	Service and repair electronic drive management systems
AURT412645A	Overhaul final drive assemblies

AUR20705 – Certificate II in Automotive Mechanical (Driveline)

Unit Number	Unit Description
AURT270278A	Use and maintain workplace tools and equipment.
AURT225667A	Use and maintain measuring equipment.
AURT271781A	Implement and monitor environmental regulations in the automotive mechanical industry.
AURC270103A	Apply safe working practices.
AURT213170A	Service final drive (driveline).
AURT207170A	Inspect and service transmissions (automatic).
AURT210170A	Inspect and service braking systems.
AURE218708A	Carry out repairs to single electrical circuits.
AURT202170A	Inspect and service cooling systems.
AURT206670A	Inspect and service transmissions (manual).
AURT212670A	Service final drive assemblies.
AURT312666A	Repair final drive assemblies.
AURT313166A	Repair final drive (driveline).
AURT412645A	Overhaul final drive assemblies.

AUR20705 – Certificate II in Automotive Mechanical (Engines)

Unit Number	Unit Description
AURT270278A	Use and maintain workplace tools and equipment.
AURT225667A	Use and maintain measuring equipment.
AURT271781A	Implement and monitor environmental regulations in the automotive mechanical industry.
AURC270103A	Apply safe working practices.
AURT202170A	Inspect and service cooling systems.
AURT201170A	Inspect and service engines
AURT200108A	Carry out servicing operations
AURT200368A	Select and use bearings, seals and gaskets
AURT203670A	Service diesel fuel systems
AURT204670A	Inspect and service emission control systems
AURC252103A	Apply basic automotive troubleshooting procedures
AURT301483A	Recondition cylinder heads
AURT301166A	Repair engines and associated engine components
AURT401145A	Overhaul engines and associated engine components (additional)
AURT321771A	Service and repair electronic compression ignition engine management systems

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AUR20705 – Certificate II in Automotive Mechanical (Hydraulics)

Unit Number	Unit Description
AURT270278A	Use and maintain workplace tools and equipment.
AURT225667A	Use and maintain measuring equipment.
AURT271781A	Implement and monitor environmental regulations in the automotive mechanical industry.
MEM5.7AB	Manual heating and thermal cutting.
AURC270103A	Apply safe working practices.
AURT409166A	Repair hydraulic systems.
AURT209170A	Service hydraulic systems.
AURT215170A	Inspect and service steering systems.
AURT316166A	Repair suspension systems.
AURT216170A	Inspect and service suspension systems.
AURT200368A	Select and use bearings, seals, gaskets, sealants and adhesives.
AURC252103A	Apply basic automotive troubleshooting procedures.
TDTD197B	Shift materials safely using manual handling methods.

AUR20705 – Certificate II in Automotive Mechanical (Machine Assembly)

Unit Number	Unit Description
AURT270278A	Use and maintain workplace tools and equipment.
AURT225667A	Use and maintain measuring equipment.
AURT271781A	Implement and monitor environmental regulations in the automotive mechanical industry.
AURE218676A	Test, service and charge batteries
AURE219331A	Install, test and repair low voltage wiring/lighting systems
AURC270103A	Apply safe working practices.
AURT213170A	Service final drive (driveline)
AURT209170A	Service hydraulic systems.
AURT215170A	Inspect and service steering systems.
AURT207170A	Inspect and service transmissions (automatic)
AURT216170A	Inspect and service suspension systems.
AURT200368A	Select and use bearings, seals, gaskets, sealants and adhesives.
AURT210170A	Inspect and service braking systems
AURT201170A	Inspect and service engines

7.0 Related Best Practices

None

8.0 Acknowledgements

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